

FROM BITANGENTS TO GENUS-2 PRYMS: THE MATH BEHIND THE PIPELINE

DOCUMENTATION NOTE

ABSTRACT. This note documents the algebraic geometry implemented in `steiner_genus2.m`, `steiner.m`, `steiner_edge.m`, `hensel_decomp.m`, and the `check_isogeny*.m` family. It explains how the 28 bitangents of a smooth plane quartic, the 63 Steiner complexes, the quartic decomposition formula $aF = L_i L_j L_k L_l + bQ^2$, and the determinantal pencil that produces genus-2 curves all fit together — and why the resulting genus-2 curves are the Prym varieties of the étale double covers of C .

Throughout, $C \subset \mathbb{P}^2$ is a smooth plane quartic over a field k of characteristic $\neq 2$, $g(C) = 3$, $K_C = \mathcal{O}_C(1)$, and $J = \text{Jac}(C)$.

1. THETA CHARACTERISTICS ON A CURVE OF GENUS 3

A *theta characteristic* on a smooth projective curve C of genus g is a line bundle $\kappa \in \text{Pic}(C)$ satisfying

$$\kappa^{\otimes 2} \cong K_C.$$

The set $\Theta(C)$ of theta characteristics is a torsor under $J[2] = \text{Pic}^0(C)[2] \cong (\mathbb{Z}/2)^{2g}$: once any κ_0 is fixed, every theta characteristic is of the form $\kappa_0 + \eta$ for a unique $\eta \in J[2]$. So $|\Theta(C)| = 2^{2g}$ — for $g = 3$, that's 64.

Each theta characteristic has a $\mathbb{Z}/2$ -valued *parity*

$$\text{par}(\kappa) := h^0(C, \kappa) \bmod 2,$$

shown by Atiyah and Mumford to be a deformation invariant [2, 8]. The number of even and odd theta characteristics is given by

$$N_+ = 2^{g-1}(2^g + 1), \quad N_- = 2^{g-1}(2^g - 1).$$

For $g = 3$: $N_+ = 36$ even and $N_- = 28$ odd. The conceptual reason for this split is the *Arf invariant* of a quadratic refinement of the Weil pairing — the topic of §2.

2. SYMPLECTIC STRUCTURE ON $J[2]$ AND THE ARF INVARIANT

2.1. The Weil pairing makes $J[2]$ into a symplectic $\mathbb{F}_2$ -space. The Weil pairing on 2-torsion,

$$e_2: J[2] \times J[2] \longrightarrow \mu_2 \cong \mathbb{F}_2,$$

is a non-degenerate alternating bilinear form. For a curve of genus g , $J[2] \cong \mathbb{F}_2^{2g}$, and $(J[2], e_2)$ admits a *symplectic basis* $e_1, f_1, \dots, e_g, f_g$ with

$$e_2(e_i, f_j) = \delta_{ij}, \quad e_2(e_i, e_j) = e_2(f_i, f_j) = 0.$$

The full symmetry group is $\text{Sp}(2g, \mathbb{F}_2)$, of order $2^{g^2} \prod_{k=1}^g (2^{2k} - 1)$. For $g = 3$ this gives $|\text{Sp}(6, \mathbb{F}_2)| = 1\,451\,520$.

Remark 2.1 ($J[2]$ vs $J/2J$ in computation). The 2-torsion subgroup $J[2]$ and the quotient $J/2J$ are abstractly isomorphic (both are $\mathbb{F}_2^{2g}$ for J finite over $\bar{k}$), but they are *Cartier dual* as Galois modules over a non-algebraically-closed field. On the Magma side, picking out the order-2 element of each cyclic factor of $J(\mathbb{F}_p)$ gives the $J[2]$ -class; reducing coordinates mod 2 gives the $J/2J$ -class. These two prescriptions agree as functions of the abstract group, but *differ* as functions of the actual Magma class group whenever $J(\mathbb{F}_p)$ has a $\mathbb{Z}/2^k\mathbb{Z}$ factor with $k \geq 2$. The pipeline systematically uses $J[2]$ (the function `ClassJ2` in `steiner_genus2.m`, which evaluates $a \mapsto (2a \operatorname{div} n) \bmod 2$ on each $\mathbb{Z}/n\mathbb{Z}$ summand with n even).

2.2. Quadratic refinements. A *quadratic refinement* of e_2 is a function $q: J[2] \rightarrow \mathbb{F}_2$ satisfying

$$q(x+y) = q(x) + q(y) + e_2(x,y) \quad \forall x, y \in J[2].$$

The set of refinements forms a torsor under the additive group of linear forms $J[2] \rightarrow \mathbb{F}_2$, which via e_2 is identified with $J[2]$ itself; in particular there are exactly 2^{2g} quadratic refinements of e_2 .

Definition 2.2 (Arf invariant). The *Arf invariant* of a quadratic refinement q is the element

$$\operatorname{Arf}(q) := \sum_{i=1}^g q(e_i) q(f_i) \in \mathbb{F}_2$$

computed in any symplectic basis (e_i, f_i) ; the sum is independent of the chosen basis.

Equivalently, the Arf invariant is the *majority bit* of q :

$$|q^{-1}(0)| = 2^{2g-1} + 2^{g-1}(-1)^{\operatorname{Arf}(q)}, \quad |q^{-1}(1)| = 2^{2g-1} - 2^{g-1}(-1)^{\operatorname{Arf}(q)}.$$

A refinement is called *even* if $\operatorname{Arf}(q) = 0$ and *odd* if $\operatorname{Arf}(q) = 1$. From the formula above, the number of even refinements is $2^{g-1}(2^g + 1)$ and the number of odd is $2^{g-1}(2^g - 1)$ — exactly N_+ and N_- from §1.

Arf introduced this invariant in [1] precisely to classify quadratic forms over $\mathbb{F}_2$. The pair $(\operatorname{rank}, \operatorname{Arf})$ is a complete invariant of a non-degenerate quadratic form over $\mathbb{F}_2$, and Arf is the unique non-trivial $\operatorname{Sp}(2g, \mathbb{F}_2)$ -invariant of a quadratic refinement of a fixed symplectic form.

2.3. Riemann–Mumford: $\Theta(C) \leftrightarrow$ refinements. The numerical coincidence between $|\Theta_{\pm}|$ and the count of even/odd refinements is no accident:

Theorem 2.3 (Riemann–Mumford). *Let C be a smooth projective curve over an algebraically closed field of characteristic $\neq 2$. There is a canonical bijection*

$$\Theta(C) \xrightarrow{\sim} \{\text{quadratic refinements of } e_2 \text{ on } J(C)[2]\}, \quad \kappa \mapsto q_{\kappa},$$

defined by

$$q_{\kappa}(\eta) = h^0(C, \kappa + \eta) + h^0(C, \kappa) \pmod{2}.$$

This bijection intertwines parity with the Arf invariant:

$$\operatorname{par}(\kappa) = h^0(C, \kappa) \bmod 2 = \operatorname{Arf}(q_{\kappa}).$$

In particular, the Arf-invariant counting formula recovers $N_{\pm} = 2^{g-1}(2^g \pm 1)$ as a *conceptual* fact, not a coincidence. References: [11, 8, 2, 5].

2.4. Translation behaviour: when does parity flip? Two consequences of Theorem 2.3 structure the rest of the paper:

Proposition 2.4 (Parity flip rule). *For $\kappa \in \Theta(C)$ and $\eta \in J[2]$,*

$$\text{par}(\kappa + \eta) - \text{par}(\kappa) = q_\kappa(\eta) \pmod{2}.$$

That is, translation by η flips parity iff η is “ q_κ -odd”.

Proposition 2.5 (Abstract even/odd dichotomy on $J[2] \setminus \{0\}$). *Fix any even theta characteristic κ_0 , and use it to identify $\Theta(C)$ with $J[2]$ via $\eta \mapsto \kappa_0 + \eta$. Then*

$$J[2] \setminus \{0\} = \underbrace{\{\eta : q_{\kappa_0}(\eta) = 0\}}_{| \cdot | = 2^{2g-1} - 1} \sqcup \underbrace{\{\eta : q_{\kappa_0}(\eta) = 1\}}_{| \cdot | = 2^{2g-1}}.$$

For $g = 3$ this is the famous decomposition

$$63 = 35 + 28,$$

where the 35 “even” classes correspond to even theta characteristics distinct from κ_0 and the 28 “odd” classes correspond to odd theta characteristics (= bitangents, by §3).

The partition into 35 and 28 is invariant under $\text{Sp}(6, \mathbb{F}_2)$ — it does *not* depend on the choice of even base κ_0 , although the labelling of individual elements does. Classically, the 35 even nonzero classes are called *syzygetic* and the 28 odd ones are called *azygetic* [12, §219]; modern accounts in [6, 5].

2.5. The Klein quartic case. For the Klein quartic, the geometric automorphism group $\text{PSL}(2, 7)$ embeds in $\text{Sp}(6, \mathbb{F}_2)$ and acts on $J[2] \setminus \{0\}$. The orbit decomposition respects the Arf decomposition $63 = 35 + 28$:

Orbit size	Arf type of η	Classical name
28	odd	azygetic / “bitangent class”
21	even	syzygetic
7 + 7 (Galois-conjugate pair)	even	syzygetic

Totals: $28 + 21 + 7 + 7 = 63$, with $35 = 21 + 7 + 7$ even and 28 odd. ✓

This is the orbit data that the Steiner-pipeline scripts process for the Klein quartic and its twist `C_twist` (see `klein_steiner_pryms.md` in the project memory for the corresponding j -invariant data).

3. BITANGENTS = ODD THETA CHARACTERISTICS

For a smooth plane quartic $C \subset \mathbb{P}^2$, the canonical bundle is $K_C = \mathcal{O}_C(1)$, so canonical divisors are cut out by lines. A *bitangent* line $L \subset \mathbb{P}^2$ to C is one whose intersection divisor on C is a sum of two double points,

$$L \cdot C = 2P + 2Q$$

(or $4P$ for a hyperflex). The line gives a section of $\mathcal{O}_C(1) = K_C$ vanishing on $2P + 2Q$, so $K_C \sim 2(P + Q)$, which means $D := P + Q$ (an effective degree-2 divisor) defines a theta characteristic

$$\kappa_L := \mathcal{O}_C(P + Q) \in \Theta(C).$$

This κ_L is *odd*: it has the obvious nonzero section vanishing on $P + Q$, so $h^0(\kappa_L) \geq 1$. By Riemann–Roch and Clifford, $h^0(\kappa_L) = 1$ generically, and $\text{Arf}(q_{\kappa_L}) = 1$.

Theorem 3.1 (Aronhold, Cayley, Salmon). *The map $L \mapsto \kappa_L$ is a bijection between bitangents to C and odd theta characteristics. In particular, a smooth plane quartic has exactly $N_- = 28$ bitangents.*

References: [12, §216], [5, §6.1].

In the code, `FindBitangentsFp` searches for lines $L = [A : B : C]$ over $\mathbb{F}_p$ such that the restriction of F to L is a perfect square as a polynomial in the line parameter. Over a number field, the script builds the *tangency ideal*

$$I_{\text{tg}} = \langle \text{coeff}_t F(t, 1, -at - b) - \text{coeff}_t (\alpha t^2 + \beta t + \gamma)^2 \rangle \subset \mathbb{Q}[a, b, \alpha, \beta, \gamma],$$

eliminates α, β, γ , and factors the resulting univariate in b (then a) to find bitangent coordinates over the splitting field. Three charts are used to cover lines through $[1:0:0]$ and the line $\{x = 0\}$.

4. STEINER COMPLEXES FROM $J[2]$

Fix two odd theta characteristics $\kappa_i, \kappa_j \in \Theta_-$ corresponding to bitangents L_i, L_j . Their difference is a 2-torsion element

$$\eta_{ij} := \kappa_i - \kappa_j \in J[2].$$

For a fixed nonzero $\eta \in J[2]$, the set

$$S_\eta := \{ \{i, j\} : \kappa_i - \kappa_j = \eta \}$$

is the *Steiner complex* associated to η .

Counting. Every unordered pair of distinct bitangents gives a unique nonzero $\eta \in J[2]$ as its difference, and there are $\binom{28}{2} = 378$ such pairs and $|J[2] \setminus \{0\}| = 63$ classes. By a uniform counting argument (the action of $J[2]$ on pairs is balanced under the symplectic geometric monodromy), each nonzero η appears the same number of times:

$$|S_\eta| = \frac{\binom{N_-}{2}}{|J[2] \setminus \{0\}|} = \frac{\binom{28}{2}}{63} = 6 \quad (g = 3).$$

So **each Steiner complex consists of exactly 6 disjoint pairs of bitangents**, i.e. 12 bitangents in total, and there are 63 such complexes. The remaining $28 - 12 = 16$ bitangents are not involved in any given S_η . Double-counting incidences via $63 \cdot 12 = 28 \cdot k$ gives $k = 27$: **each bitangent appears in exactly 27 Steiner complexes** (one per partner).

References: [5, §6.1.5], [12, §219], [6].

The number $63 \cdot 6 = 378 = \binom{28}{2}$ accounts for every unordered pair of bitangents. The combinatorics encodes the symmetric structure of the *Cayley quartic* or “Aronhold set”: modulo $\text{Sp}(6, \mathbb{F}_2)$, this is the orbit data of the natural action on the 28 odd characteristics.

4.1. Computing η_{ij} in Magma: the $J[2]$ class, not $J/2J$. The function field of C over $\mathbb{F}_p$ is constructed in `steiner_genus2.m`. For each bitangent line L_i , the script forms the function $L_i/L_{\text{ref}} \in k(C)$, takes

$$\frac{1}{2} \text{div}(L_i/L_{\text{ref}}) \in \text{Pic}^0(C),$$

which is well-defined exactly because L_i is bitangent (all valuations even), and recovers the corresponding $J[2]$ class $\eta_i = \kappa_i - \kappa_{\text{ref}}$.

The translation from a class group element to a $J[2]$ element is performed by the function `ClassJ2`:

```
j2 := [];
for k in [1..#invs] do
  if invs[k] eq 0 then continue; end if;
  if invs[k] mod 2 ne 0 then continue; end if;
  // For Z/nZ (n even), the unique 2-torsion element is n/2.
  // The map a -> (2*a div n) mod 2 sends 0 -> 0 and n/2 -> 1.
  Append(~j2, (2 * coords[k] div invs[k]) mod 2);
end for;
```

This is *not* the same as `ReduceMod2`, which reduces every coordinate of the class-group element mod 2 — that would give the $J/2J$ class. Cf. Remark 2.1; the Steiner pipeline needs the literal 2-torsion line bundle, so `ClassJ2` is the correct primitive.

Pairs $\{i, j\}$ are then binned by the $J[2]$ class $\eta_i - \eta_j$ to recover the 63 complexes (each with 6 pairs, verified by the assertions `nclasses eq 63` and `s eq 6`).

4.2. Even vs. odd η , and the parity of the complex. By the Arf decomposition of §2, the quadratic refinement q_{κ_0} on $J[2]$ partitions the nonzero classes into 35 syzygetic (even, $q = 0$) and 28 azygetic (odd, $q = 1$). The classical Steiner-complex theory of plane quartics is developed for the syzygetic case: a syzygetic Steiner complex is the data needed to write the quartic decomposition $aF = L_i L_j L_k L_l + bQ^2$ of §5.

For a plane quartic with $g = 3$, the 28 odd nonzero η are in canonical bijection with the 28 bitangents (via the parity-flip rule: the “bitangent class” of a bitangent L_i is $\kappa_i - \kappa_0$). The 35 even nonzero η correspond to the 35 even theta characteristics of the form $\kappa_0 + \eta$; equivalently, to Cayley octads / Aronhold systems of seven bitangents [5, Ch. 6].

For the Klein quartic specifically, the orbit decomposition $28 + 21 + 7 + 7$ under $\mathrm{PSL}(2, 7)$ refines this: one orbit of size 28 = the azygetic orbit, three orbits (sizes 21, 7, 7) inside the 35 syzygetic classes.

5. THE QUARTIC DECOMPOSITION FORMULA

Fix a Steiner complex $S_\eta = \{\{i_1, j_1\}, \dots, \{i_6, j_6\}\}$ and pick *two* of its six pairs, say $\{i, j\}$ and $\{k, l\}$. The classical identity is:

Theorem 5.1. *There exists a conic $Q \in H^0(\mathbb{P}^2, \mathcal{O}(2))$ and scalars $a, b \in k$ (not both zero) such that*

$$(1) \quad aF = L_i L_j L_k L_l + bQ^2.$$

The conic Q is the unique (up to scalar) contact conic that is tangent to C at the eight contact points $\{P_i, P_j, P_k, P_l\}$ of the four bitangents.

References: [12, §220], [5, Prop. 6.1.7], and [4] for a modern reconstruction perspective.

The geometric content of (1) is: cutting out $Q^2 = 0$ on C gives the divisor $2(P_i + P_j + P_k + P_l)$, which equals the divisor cut by $L_i L_j L_k L_l$ on C (each line contributes $2P_m$). Both sides therefore give effective representatives of $4K_C$, and (1) is the unique linear relation between $L_i L_j L_k L_l$, Q^2 , and F in degree 4.

In the code, two linear-algebra steps implement this:

- (1) `FindConic` assembles a 12×10 matrix encoding the conditions “the restriction of Q to each L_m is proportional to the contact quadratic h_m of L_m ” for $m \in \{i, j, k, l\}$ (3 conditions per line, so 12 rows; 6 unknowns for Q and 4 proportionality constants λ_m , so 10 columns). The kernel is 1-dimensional and gives Q .
- (2) The quartic-decomposition step then assembles the 3×15 matrix whose rows are the coefficient vectors of F , $L_i L_j L_k L_l$, and Q^2 in the basis of degree-4 monomials, and reads off the relation $aF + (-1)L_i L_j L_k L_l + bQ^2 = 0$ from the kernel.

For each Steiner complex there are $\binom{6}{2} = 15$ pair-of-pairs, hence $63 \cdot 15 = 945$ such conics in total. The script verifies them all and prints `Verified 945 / 945`.

6. FROM THE DECOMPOSITION TO A 3×3 SYMMETRIC PENCIL

Rewrite (1) as

$$bQ^2 - (-L_i L_j)(L_k L_l) = aF.$$

Set $Q_1 := b L_i L_j$, $Q_2 := b Q$, $Q_3 := -L_k L_l$ (each a homogeneous quadratic form, i.e. an element of $\text{Sym}^2 V^*$ with $V = k^3$). Then

$$Q_1 Q_3 - Q_2^2 = -ab F,$$

i.e. C is cut out (up to scalar) by the *discriminant* of the symmetric matrix

$$M := \begin{pmatrix} Q_1 & Q_2 \\ Q_2 & Q_3 \end{pmatrix},$$

viewing Q_1, Q_2, Q_3 as quadratic forms on $\mathbb{P}^2$. Equivalently, C is the *degeneracy locus* of the pencil $\{Q_1 + 2tQ_2 + t^2Q_3\}$ of conics, parametrised by $t \in \mathbb{P}^1$.

Now upgrade to the level of 3×3 symmetric matrices on $\mathbb{P}^2$ itself: write each conic Q_m as a symmetric matrix $M_m \in \text{Sym}^2(k^3)$ of size 3, and form the matrix-valued polynomial

$$M(t) := M_1 + 2t M_2 + t^2 M_3 \in \text{Mat}_3(k[t]).$$

The locus

$$\tilde{C} := \{ (t, [v]) \in \mathbb{P}^1 \times \mathbb{P}^2 : v^\top M(t) v = 0 \}$$

is a $(2, 2)$ -divisor in $\mathbb{P}^1 \times \mathbb{P}^2$; the projection $\tilde{C} \rightarrow \mathbb{P}^1$ is a conic bundle (smooth conics for generic t , degenerating to line pairs when $\det M(t) = 0$).

Definition 6.1. Set $B := \{ t \in \mathbb{P}^1 : \det M(t) = 0 \}$. For generic input, $\det M(t)$ is a polynomial of degree 6, so $|B| = 6$. The hyperelliptic curve

$$\mathcal{H} : y^2 = -\det M(t)$$

has 6 Weierstrass points (or 5+ the point at infinity when $\det$ has degree 5), hence genus

$$g(\mathcal{H}) = \frac{6-2}{2} = 2.$$

This is the function `MakeGenus2` in `steiner_genus2.m` and `steiner.m`.

7. WHY $\mathcal{H}$ IS THE PRYM VARIETY

For each non-zero $\eta \in J(C)[2]$ there is an associated *étale double cover* $\pi : D_\eta \rightarrow C$, classified by $\eta \in H^1(C, \mathbb{Z}/2) \cong J(C)[2]$. By Riemann–Hurwitz,

$$2(g_{D_\eta} - 1) = 2 \cdot 2(g_C - 1) = 8 \implies g_{D_\eta} = 5.$$

The covering involution σ acts on $\text{Jac}(D_\eta)$, and the *Prym variety* is the connected component of the kernel of the trace map,

$$P(D_\eta/C) := (\ker \text{Nm}_\pi)^0 \subset \text{Jac}(D_\eta).$$

It is a $(g_{D_\eta} - g_C) = 2$ -dimensional principally polarized abelian variety (the principal polarization is half the restriction of the canonical polarization on $\text{Jac}(D_\eta)$, the special fact about étale double covers due to Mumford).

References: [9, 3, 7].

For a generic curve of genus ≤ 5 , the Prym map $\mathcal{R}_g \rightarrow \mathcal{A}_{g-1}$ is dominant; for $g = 3$ the source has dimension $\dim \mathcal{M}_3 + 6 = 12$ and the target $\mathcal{A}_2$ has dimension 3, so generic fibres have dimension 9.

Theorem 7.1 (Wirtinger / Recillas / Verra / Dolgachev). *Let $C \subset \mathbb{P}^2$ be a smooth plane quartic, $\eta \in J(C)[2] \setminus \{0\}$, and $\pi : D_\eta \rightarrow C$ the corresponding étale double cover. Then $P(D_\eta/C)$ is the Jacobian of a smooth genus-2 curve $\mathcal{H}_\eta$, and $\mathcal{H}_\eta$ can be realised explicitly as the discriminant curve $y^2 = -\det M(t)$ of any pencil $M(t)$ of 3×3 symmetric matrices constructed from any pair-of-pairs in the Steiner complex S_η .*

References: [14, 10, 3, 13, 5].

In other words: *the genus-2 curves produced by `steiner.m` and `steiner_genus2.m` are exactly the 63 Prym varieties* of the étale double covers of C , presented as Jacobians of explicit genus-2 curves. Different choices of pair-of-pairs within the same Steiner complex give the same genus-2 curve (up to isomorphism), reflecting the well-definedness of the Prym.

8. SPLITTING THE PRYM: ELLIPTIC FACTORS VIA THE $\mathbb{Z}/3$ TRICK

A 2-dimensional ppav A is either irreducible or $(2, 2)$ -isogenous to a product $E_1 \times E_2$ of elliptic curves. When $A = J(\mathcal{H})$ for a genus-2 hyperelliptic curve, one tests for an elliptic decomposition by looking for an order-3 automorphism of $\mathbb{P}^1$ that *permutes the 6 Weierstrass points* of $\mathcal{H}$.

Concretely: an order-3 Möbius transformation has two fixed points and partitions the remaining points into orbits of size 3. If the 6 Weierstrass points $\{p_1, \dots, p_6\}$ split as a single $\mathbb{Z}/3$ -orbit of length 3 plus another orbit of length 3 (no fixed Weierstrass points), then this $\mathbb{Z}/3$ action descends to a degree-3 map $\mathcal{H} \rightarrow E$ onto an elliptic curve, and the second elliptic factor E' is the quotient by the genus-2 hyperelliptic involution.

In the script `FindOrbit`, this is implemented as: for each ordered triple (p_a, p_b, p_c) of Weierstrass points, apply the unique Möbius transformation sending $(p_a, p_b, p_c) \mapsto (0, 1, \infty)$, then check whether the images of the remaining three points form a $\mathbb{Z}/3$ -orbit $\{x, 1 - 1/x, 1/(1 - x)\}$ — these are the cyclic permutations of the generic cross-ratio under the order-3 element of the anharmonic group $S_3 \subset \mathrm{PGL}_2$ acting on cross-ratios.

When such an x is found, the cross-ratio $\lambda = x$ identifies the elliptic curve via the standard formula

$$j(E) = 256 \cdot \frac{(1 - s(1 - s))^3}{s^2(1 - s)^2}, \quad s = (1 - \lambda)(\lambda + \sqrt{\lambda^2 - \lambda + 1})^2.$$

(The intermediate variable s comes from a 2-isogeny normalisation — with the square root, the formula returns the j -invariant of one of the two isogeny factors of the $(2, 2)$ -decomposition.)

For the Klein-twist case `C_twist`, this returns the j -invariants documented in `check_isogeny*.m` and in `klein_steiner_pryms.md`:

PSL(2, 7) orbit	Arf	min poly of j over $\mathbb{Q}$	splitting
size 28 (azygetic, “bitangent class”)	odd	$v^2 + 13856v - 26578688$	$\mathbb{Q}(\sqrt{7})$, $j = -6928 \pm 3264\sqrt{7}$
size 7 + 7 (syzygetic, conjugate pair)	even	$v^4 - 103439v^3 + 6670405329v^2 + \dots$	irreducible/ $\mathbb{Q}$, splits as $\mathrm{mp}_2 \cdot \mathrm{mp}_3$ over $\mathbb{Q}(\sqrt{-7})$
size 21 (syzygetic)	even	not separately recorded	re-derive from <code>steiner.m</code>

The script `check_isogeny_correct.m` then proves rigorously, via the CM-field criterion (the squarefree part of $a_p^2 - 4p$ for ordinary reduction), that the elliptic factors from different PSL(2, 7) orbits are not $\overline{\mathbb{Q}}$ -isogenous, and that none of them is the curve $E = 49a1$ that appears in $J(C_{\mathrm{twist}}) \cong E^3$.

9. END-TO-END PIPELINE SUMMARY

Given a smooth plane quartic $F(x, y, z) \in k[x, y, z]_4$:

Step	Object	Code	Math
0	reduce mod good prime p	search loop	smoothness check
1	28 bitangents over $\mathbb{F}_p$ (later over $K \supset \mathbb{Q}$)	FindBitangentsFp , tangency ideal + elimination	odd theta characteristics §3
2	63 Steiner complexes	function-field $J[2]$ classes of L_i/L_{ref} , via ClassJ2	Steiner complexes §4
3	witnessing conic Q for each pair-of-pairs	12×10 linear system (FindConic)	contact conic §5
4	scalars a, b in $aF = L_i L_j L_k L_l + bQ^2$	nullspace of 3×15 coefficient matrix	quartic decomposition §5
5	symmetric pencil $M(t) = M_1 + 2tM_2 + t^2M_3$	MakeGenus2	linear system of conics §6
6	genus-2 curve $\mathcal{H} : y^2 = -\det M(t)$	HyperellipticCurve	Prym = Jacobian §7
7	Igusa invariants and (when $(2, 2)$ -reducible) elliptic factor j -invariants	IgusaInvariants , FindOrbit	$\mathbb{Z}/3$ on Weierstrass points §8
8	rigorous (non-)isogeny among elliptic factors	check_isogeny_correct	CM-field criterion (squarefree part of $a_p^2 - 4p$)

The general-purpose drivers **steiner_genus2.m** and **hensel_decomp.m** accept any smooth plane quartic over $\mathbb{Q}$ as input. **steiner_edge.m** is hardcoded to the Edge quartic $25(x^4 + y^4 + z^4) - 34(x^2y^2 + x^2z^2 + y^2z^2)$ (an S_4 -symmetric quartic with all 28 bitangents real over $\mathbb{Q}$); **steiner.m** is the legacy predecessor hardcoded to **C_twist**.

REFERENCES

- [1] C. Arf, *Untersuchungen über quadratische Formen in Körpern der Charakteristik 2 (Teil I)*, J. Reine Angew. Math. **183** (1941), 148–167.
- [2] M. F. Atiyah, *Riemann surfaces and spin structures*, Ann. Sci. École Norm. Sup. (4) **4** (1971), 47–62.
- [3] A. Beauville, *Variétés de Prym et jacobiniennes intermédiaires*, Ann. Sci. École Norm. Sup. (4) **10** (1977), 309–391.
- [4] L. Caporaso and E. Sernesi, *Recovering plane curves from their bitangents*, J. Algebraic Geom. **12** (2003), 225–244.
- [5] I. Dolgachev, *Classical Algebraic Geometry: A Modern View*, Cambridge Univ. Press, 2012.
- [6] I. Dolgachev and D. Ortland, *Point sets in projective spaces and theta functions*, Astérisque **165** (1988).
- [7] R. Donagi, *The fibers of the Prym map*, in: Curves, Jacobians, and Abelian Varieties (Amherst, MA, 1990), Contemp. Math. **136**, AMS, 1992, pp. 55–125.
- [8] D. Mumford, *Theta characteristics of an algebraic curve*, Ann. Sci. École Norm. Sup. (4) **4** (1971), 181–192.
- [9] D. Mumford, *Prym varieties I*, in: *Contributions to Analysis* (L. Ahlfors et al., eds.), Academic Press, 1974, pp. 325–350.
- [10] S. Recillas, *Jacobians of curves with g_4^1 ’s are the Pryms of trigonal curves*, Bol. Soc. Mat. Mexicana (2) **19** (1974), 9–13.
- [11] B. Riemann, *Theorie der Abel’schen Functionen*, J. Reine Angew. Math. **54** (1857), 115–155; reprinted in *Gesammelte Mathematische Werke* (1876).
- [12] G. Salmon, *A Treatise on the Higher Plane Curves*, 3rd ed., Hodges, Foster, and Figgis, 1879.
- [13] A. Verra, *The fibre of the Prym map in genus three*, Math. Ann. **276** (1987), 433–448.
- [14] W. Wirtinger, *Untersuchungen über Thetafunctionen*, Teubner, Leipzig, 1895.